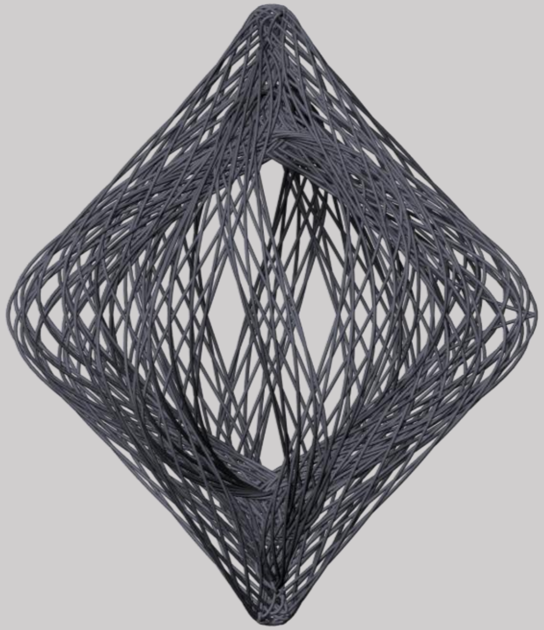


portfolio

NIMA ZAHIRI

NIMA ZAHIRI

Civil Eng., Ph.D. Timber Construction

Email: zahiri.nima@gmail.com
Phone: +47 459 15 627
LinkedIn: linkedin.com/in/nimazahiri/
Web-page: nimazahiri.github.io
Bernau bei Berlin, Germany

QUALIFICATIONS

International work experience
Top-tier academic education
Technical professional portfolio
Broad software expertise
Solid production competence
Multilingual communication skills

ACADEMIC EDUCATION

Ph.D. Timber Construction
2023 **Computational Design**
2020 **and Automated Production**
NTNU Gjøvik, NO

M.Sc. ITECH
2019 **Integrative Technologies and**
2017 **Architectural Design Research**
University of Stuttgart, DE

B.Sc. Civil Engineering
2011 **Structural and**
2006 **Building Construction**
Razi University, IR

SOFTWARE EXPERTISE

Parametric Design – Grasshopper
Programming – Python, C#
BIM Modeling – Revit
FE Simulation – Karamba3D, Fusion 360
Timber Construction – cadwork, SEMA
Fabrication – Hundegger Cambium

WORK EXPERIENCE

Timber Construction Designer
Now. **Design and Planning of**
2023 **Modular Timber Construction**
timpla GmbH, Eberswalde, DE

Designer / 3D Specialist
2020 **Computational Design of**
Complex Concrete Reinforcement
Werner Sobek AG, Stuttgart, DE

Research Assistant
2019 **Research Support in**
2017 **Doctoral Projects**
University of Stuttgart, DE

Project Engineer
2016 **Building Information Modelling**
2014 **And Construction Detailing**
T.AHA Architecture, Tehran, IR

LANGUAGE PROFICIENCY

Persian – native
English – fluent at an academic level
German – advanced working proficiency
Norwegian – independent communication

This portfolio focuses on computational design, integrated technologies and digital fabrication.

It consists of two chapters:

CHAPTER I

tech

technology projects

CHAPTER II

wood

woodbased projects

Please see in full screen

CHAPTER I

tech

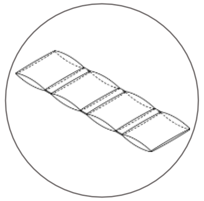
technology
projects

The first chapter focuses on **integrative technologies** in architecture, in the context of computational design and digital fabrication.

CHAPTER I

technology
projects

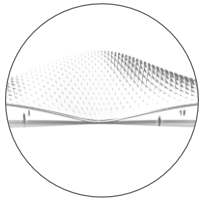
1 **FUTURE TRADITION**
Robotic Clay Printing



2 **JOINT FREE MOVEMENT**
Fiber Composite Production

3 **T-BEAM**
Bending Active Plates

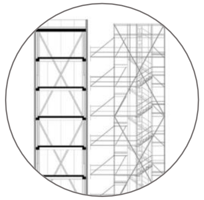
4 **FLECTOFOLD**
Frame Steel CNC Fabrication



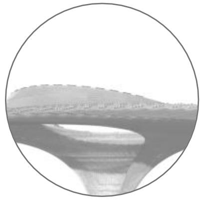
5 **TRAIN STATION**
Flectofold Competition Proposal

6 **ITECH RESEARCH DEMONSTRATOR 2019**
Biomimetic Adaptive Architecture

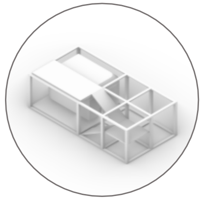
7 **ELECTROACTIVE SKIN**
Electroactive - Textile Production



8 **ADAPTIVE TOWER**
SFB1244 Demonstrator



9 **MAIN STATION STUTTGART 21**
Complex Concrete Reinforcement



10 **MODUS HIGH-RISES**
Modular Procedural Assembly

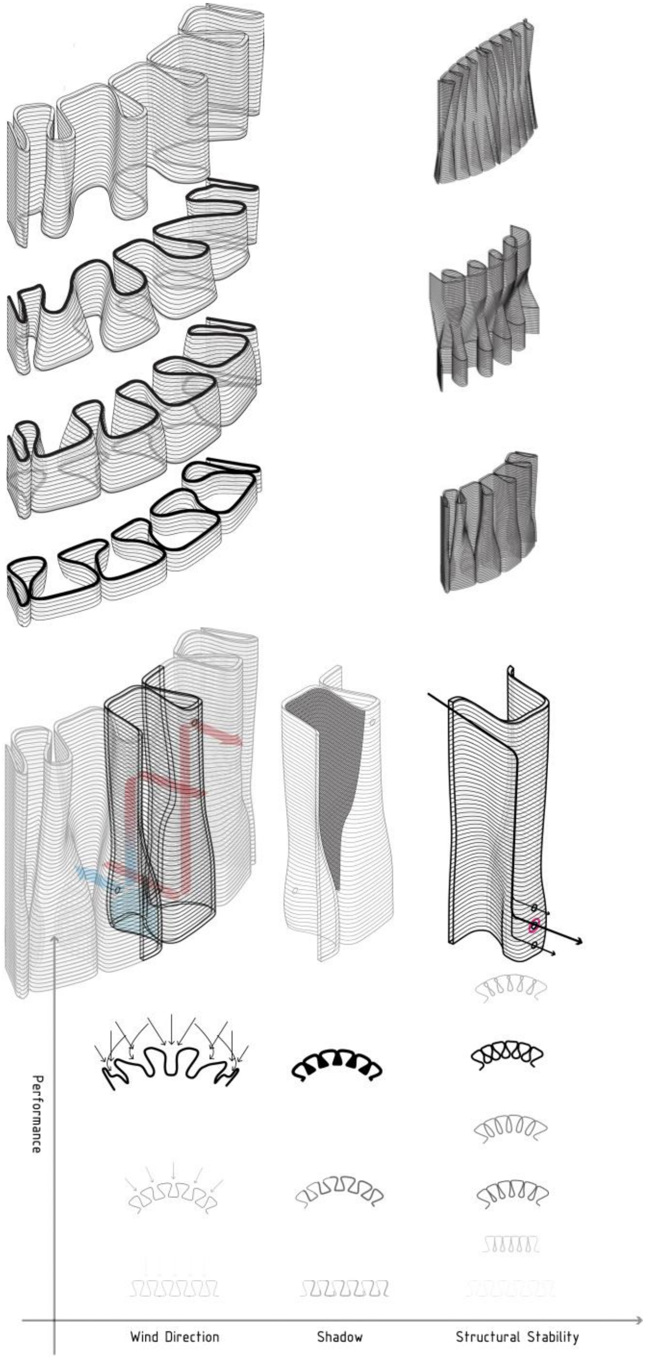
work **FUTURE TRADITION**
Rethink of Vernacular Clay Cooling System

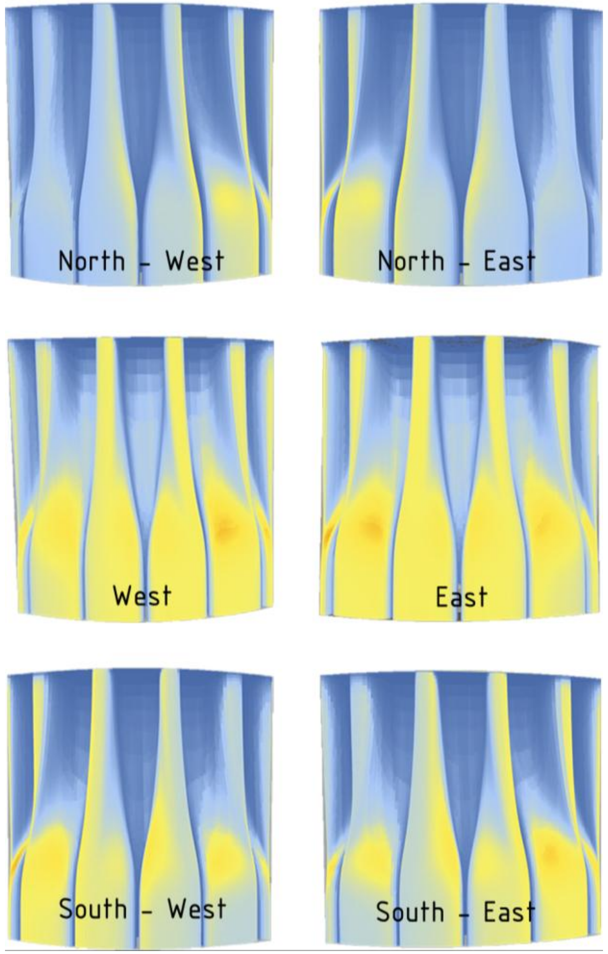
tech KUKA Robotic Clay Printing, Environmental Analysis

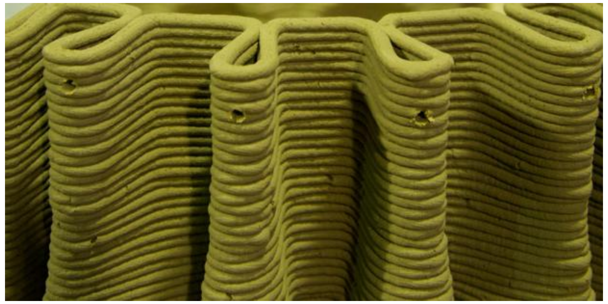
role Competition: Future Tradition
2016 *Contemporary Architect Association (CAA), IR*

task Entire Process

The main aim of the project is to reinterpret the environmental mechanism of traditional clay vase in the context of architecture, using robotic fabrication.







work **JOINT FREE MOVEMENT**
Shape-changing Fiber Composite Plates

tech Pneumatic Cushions, Glass Fiber

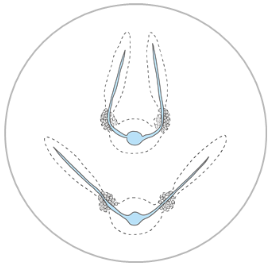
role ITECH Course: Architectural Biomimetic
2017 *ICD Institute, University of Stuttgart, DE*

task Entire Process

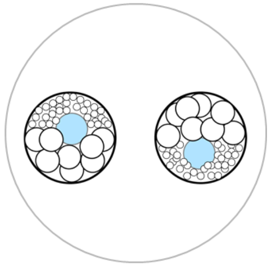
The aim of this research project is to control the elastic stiffness and bending behavior of a fiber composite plate with pneumatic actuation.

Biological Role Model: Mimosa Pudica

One branch

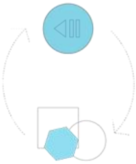


Petiole with leaves



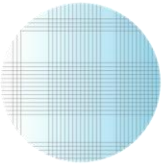
Cells in Pulvini

RELEVANT PRINCIPLES FOR ABSTRACTION



Fusion of actuators

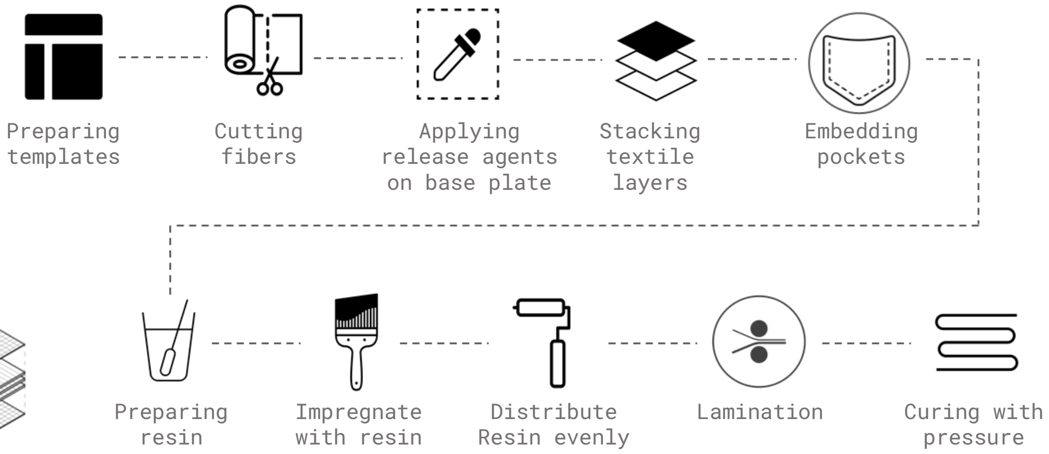
Pressure change

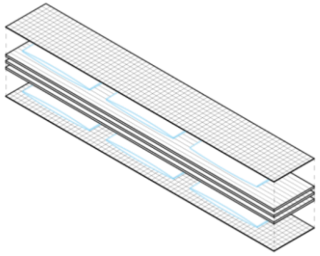


Material gradient

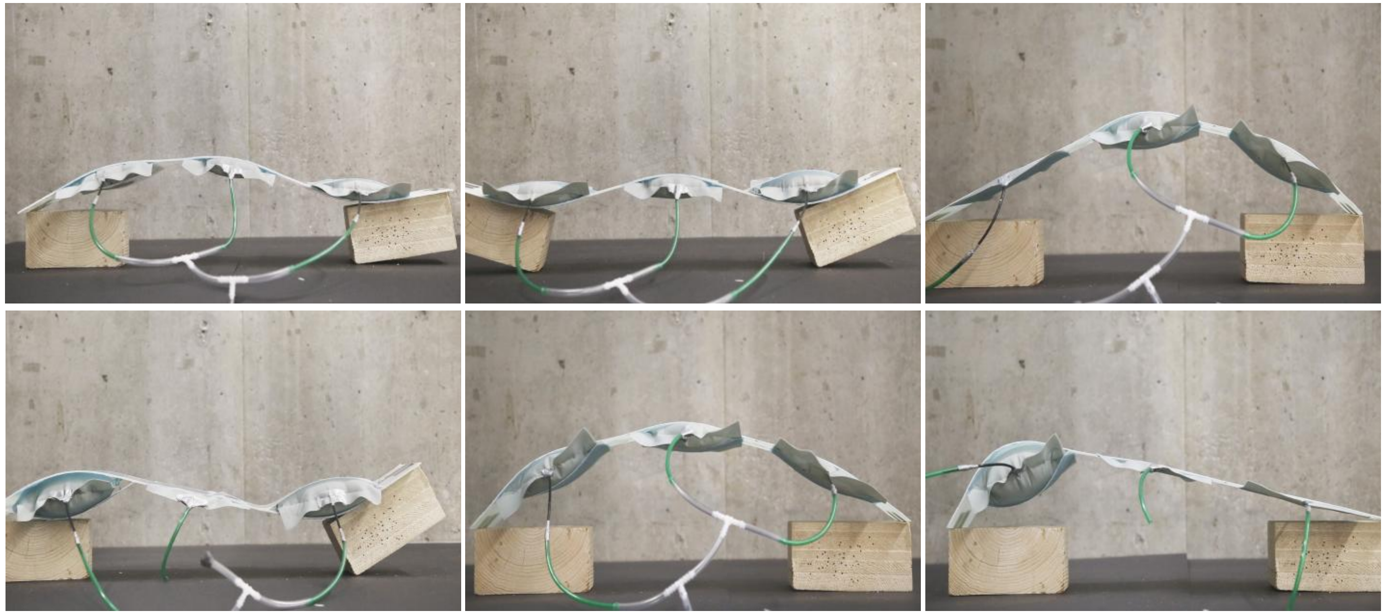
Compliance

Reversibility and freedom





1x Glass fiber woven fabric 0/60
2x Plastic (pocket)
3x Glass fiber fabric, unidirectional
2x Plastic (pocket)
1x Glass fiber woven fabric 0/60

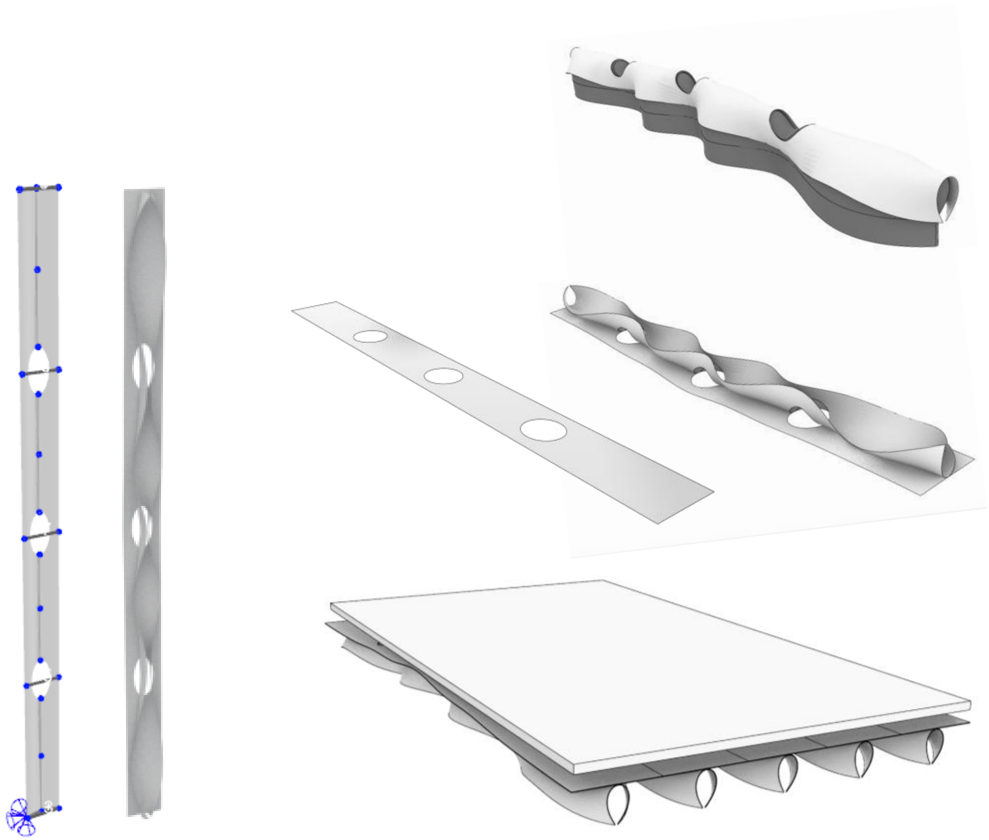


work **T-BEAM**
Curved Bending T Cross-sections

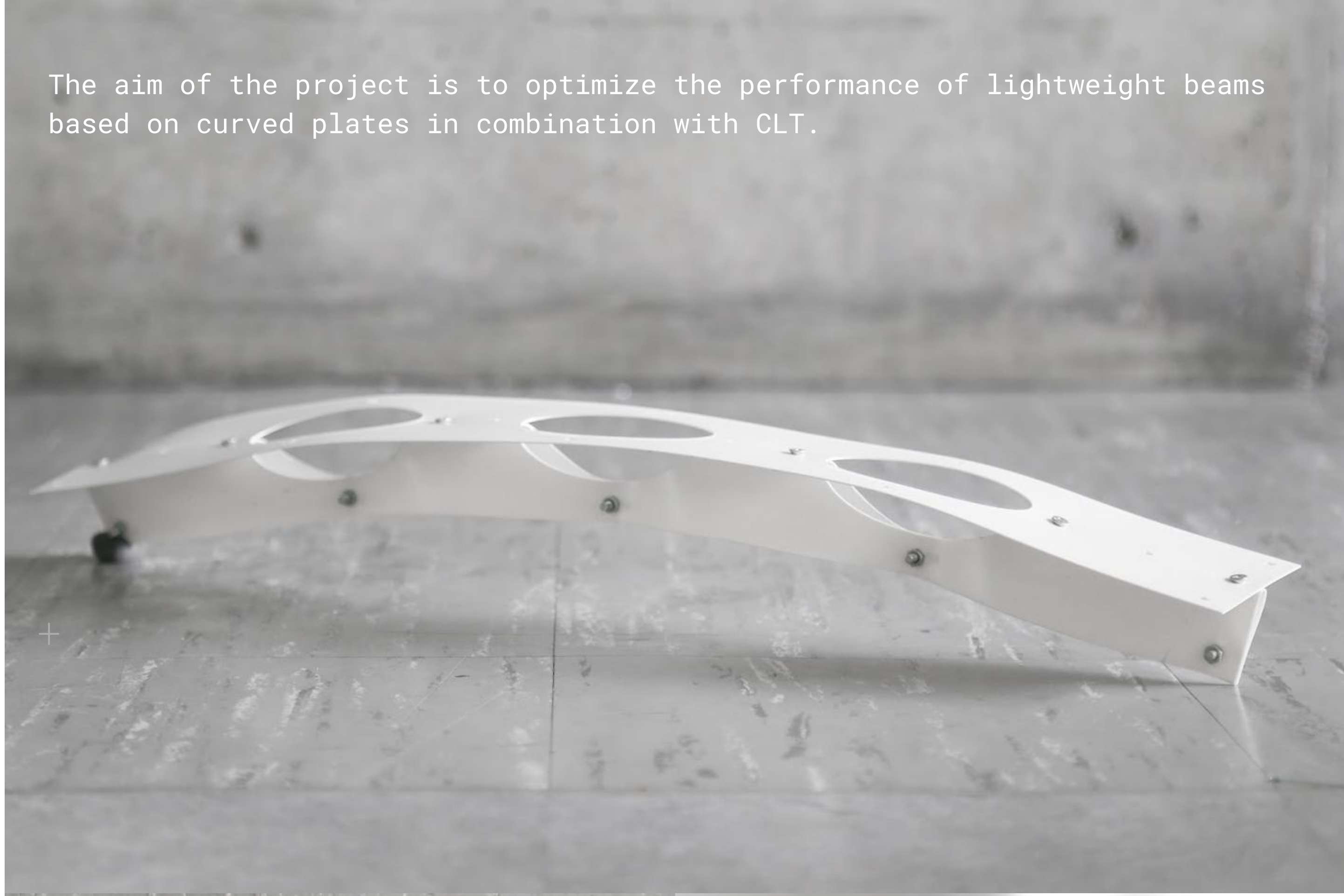
tech Typology Optimization and Manual Prototyping

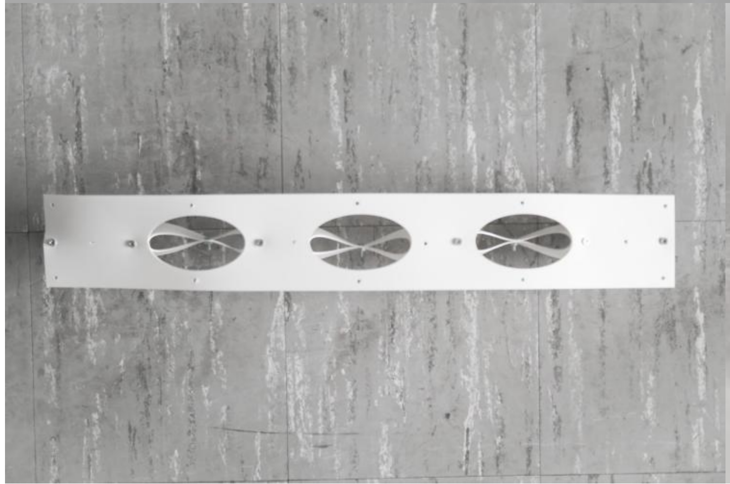
role ITECH Course: Material and Structure
2018 *ITKE Institute, University of Stuttgart, DE*

task Entire Process



The aim of the project is to optimize the performance of lightweight beams based on curved plates in combination with CLT.



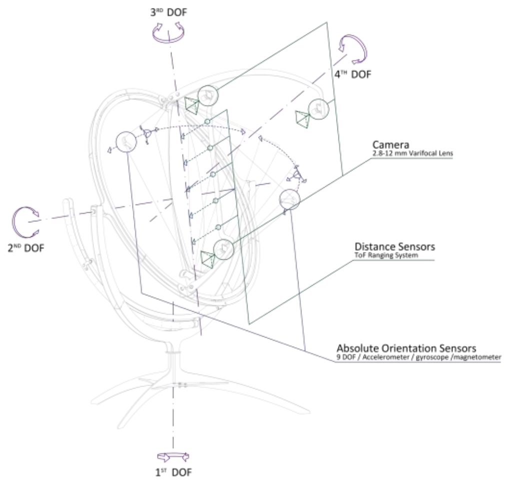


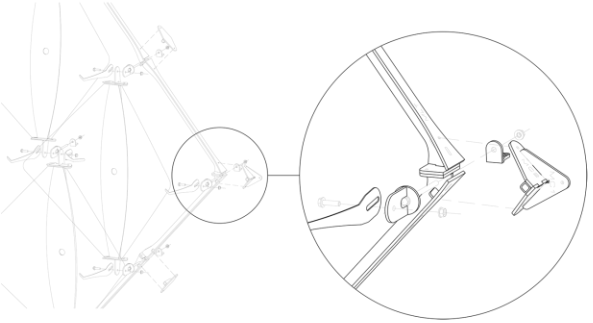
work **FLECTOFOLD**
Demonstrator & Performance Monitoring Device

tech Compliant Mechanism

role Student Assistant (HiWi)
2018 *ITKE Institute, University of Stuttgart, DE*

task Design Development, Fabrication, Assembly





The aim of this project was to monitor, evaluate and showcase the potential of materially graduated FRP as a material solution for climate adaptive building envelopes. To read more, click here.

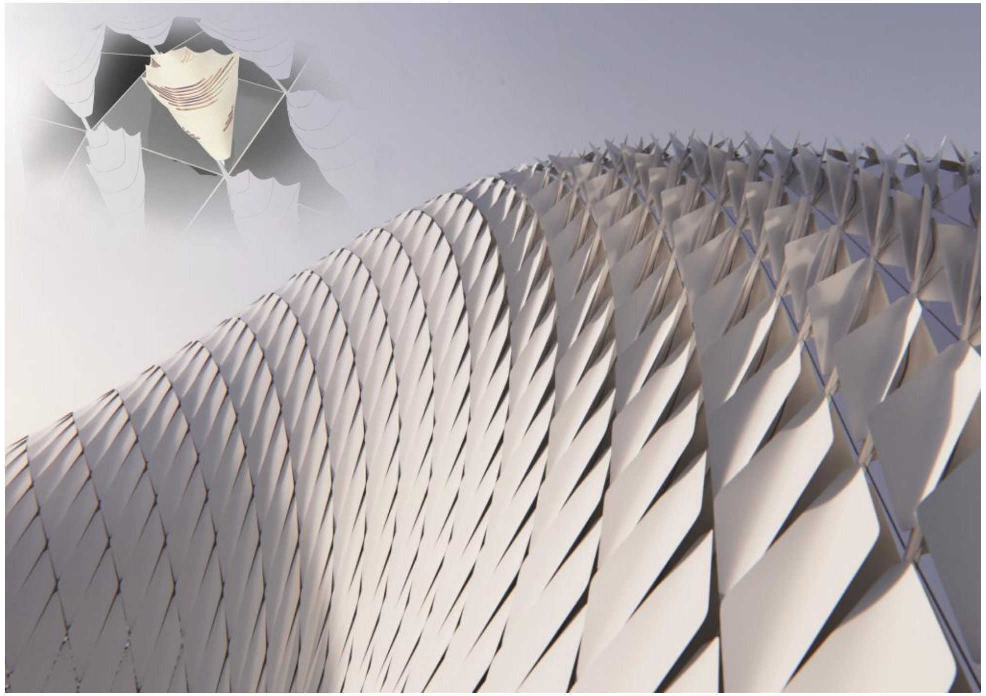
work **TRAIN STATION**
FlectoFold Design Development

tech Compliant Mechanism

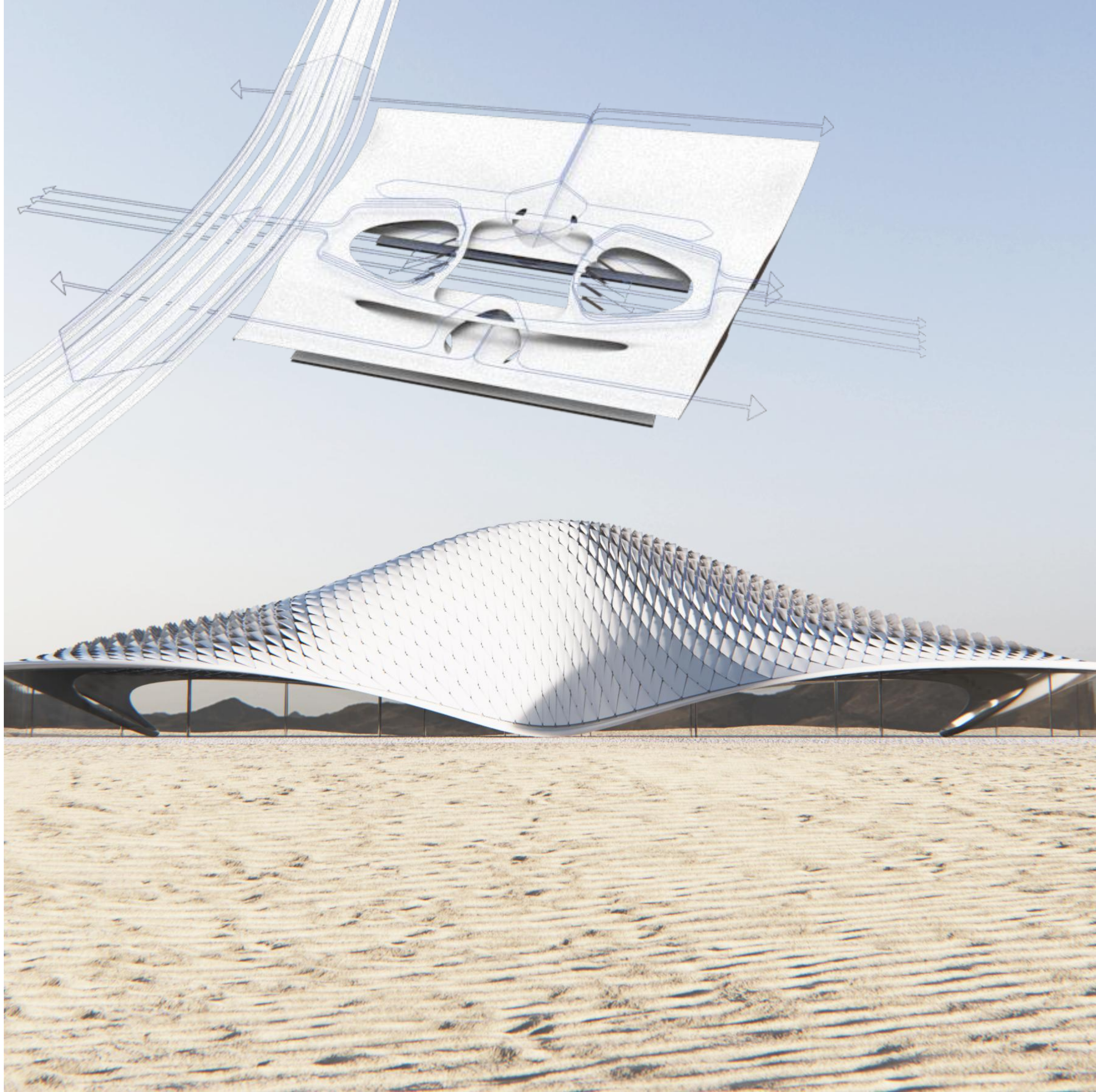
role Competition
2018 Saman Saffarian Architecture Office, CZ

task Circulation Design and Functioning Diagrams

The aim of this project was to propose an airport using segmented Flectofold shell for environmental adaptation.



Pictures © Saman Saffarian

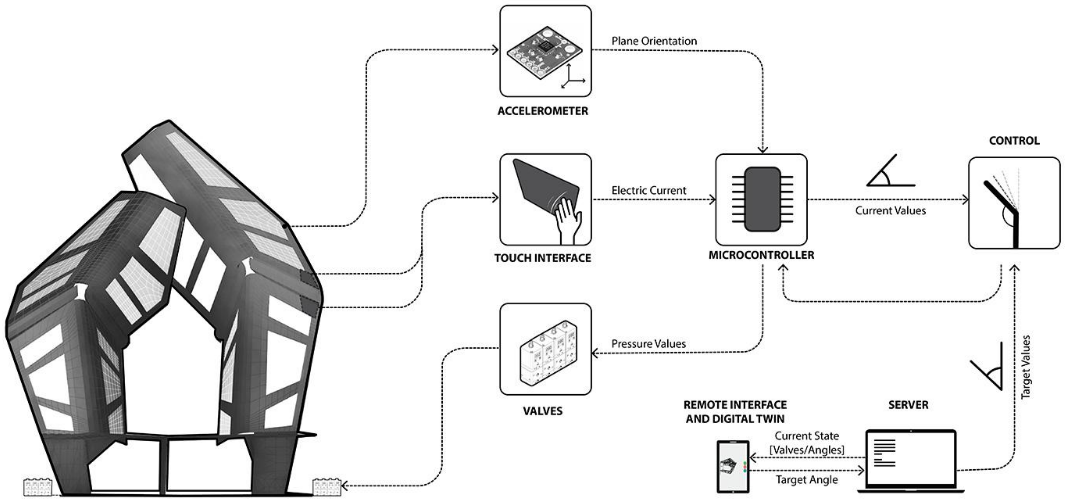


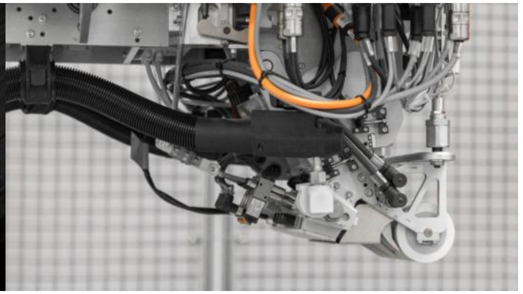
work **ITECH RESEARCH DEMONSTRATOR 2018-19**
Biomimetic Adaptive Architecture

tech Robotic Carbon Fiber Placement, Haptic Adaptive

role ITECH Studio Project (2019)
2019 *ICD / ITKE Institutes, University of Stuttgart, DE*

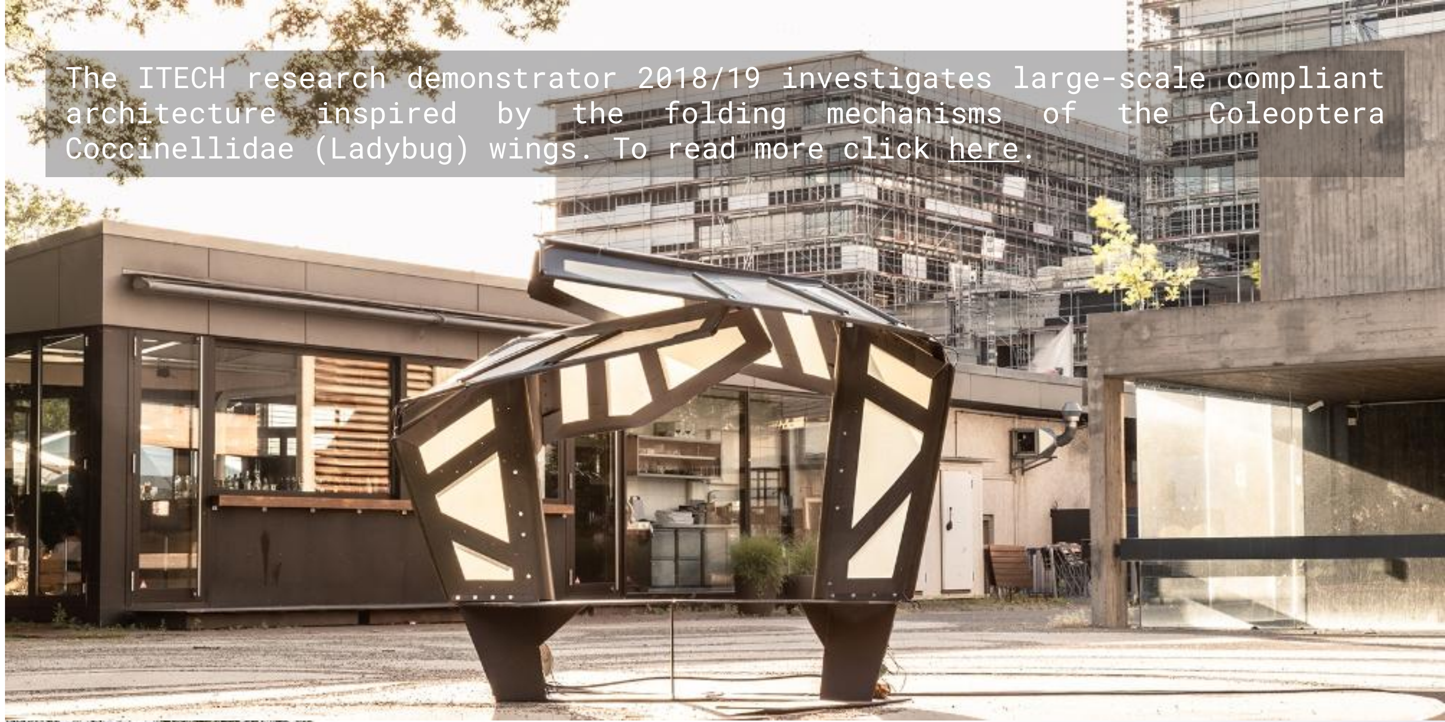
task Simulation, Fabrication





Pictures © ITKE / ICD University of Stuttgart

The ITECH research demonstrator 2018/19 investigates large-scale compliant architecture inspired by the folding mechanisms of the Coleoptera Coccinellidae (Ladybug) wings. To read more click here.



work **ELECTROACTIVE SKIN**
Towards Bio-inspired Responsive Envelopes

tech Integration of EAP with Textile

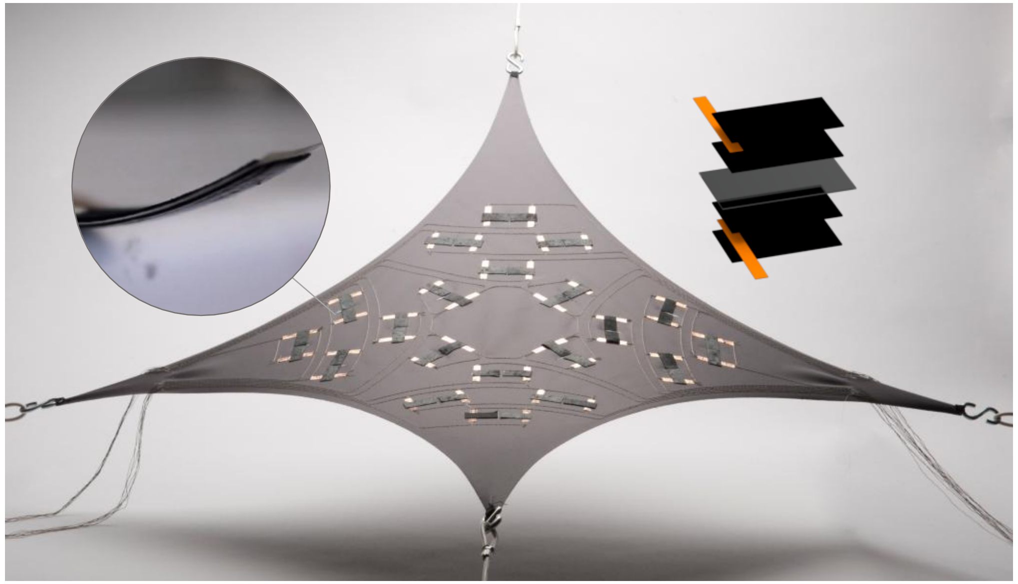
role Master’s Thesis (2019)
2019 *University of Stuttgart (BioMat/ITKE, ICD), DE*
Fraunhofer Institute (IPA), DE
Institute of Aircraft Design (IFB), DE

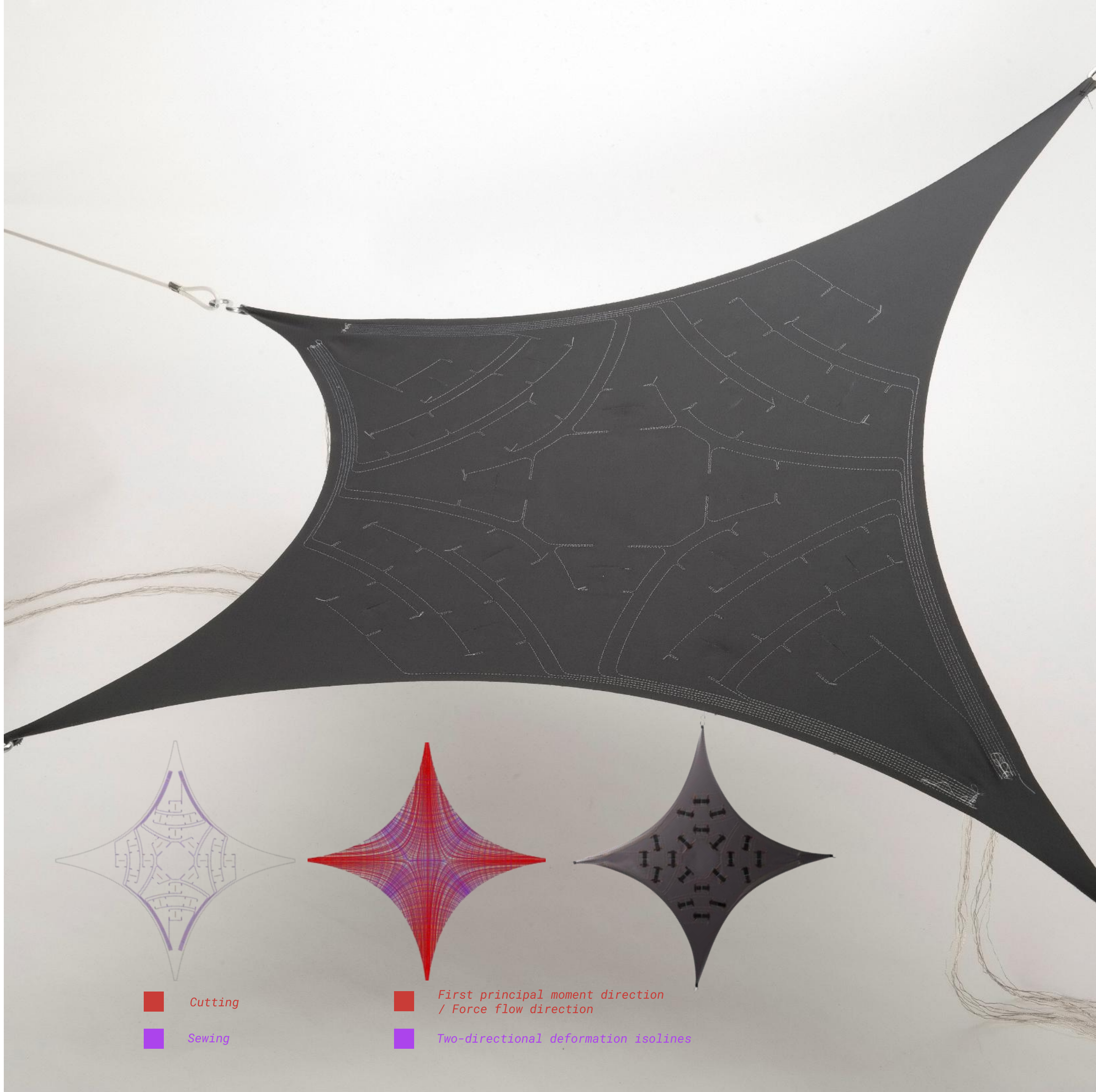
task Entire Process

The aim of this research is to investigate actuatable apertures in architectural skin through incorporation of ionic electroactive polymer.

To see the video, click here.

To read the published paper, click here.





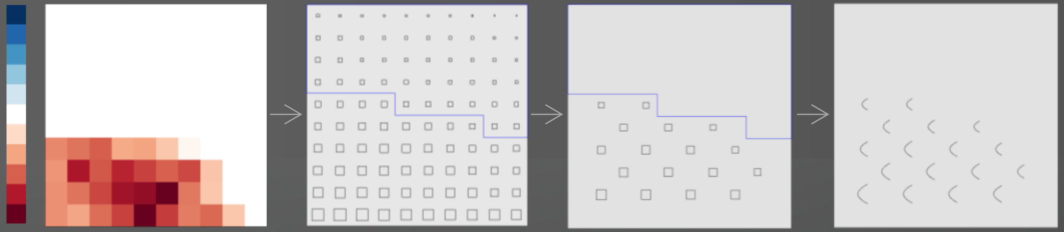
work **ADAPTIVE TOWER**
SFB1244 Adaptive Building Skin Demonstrator

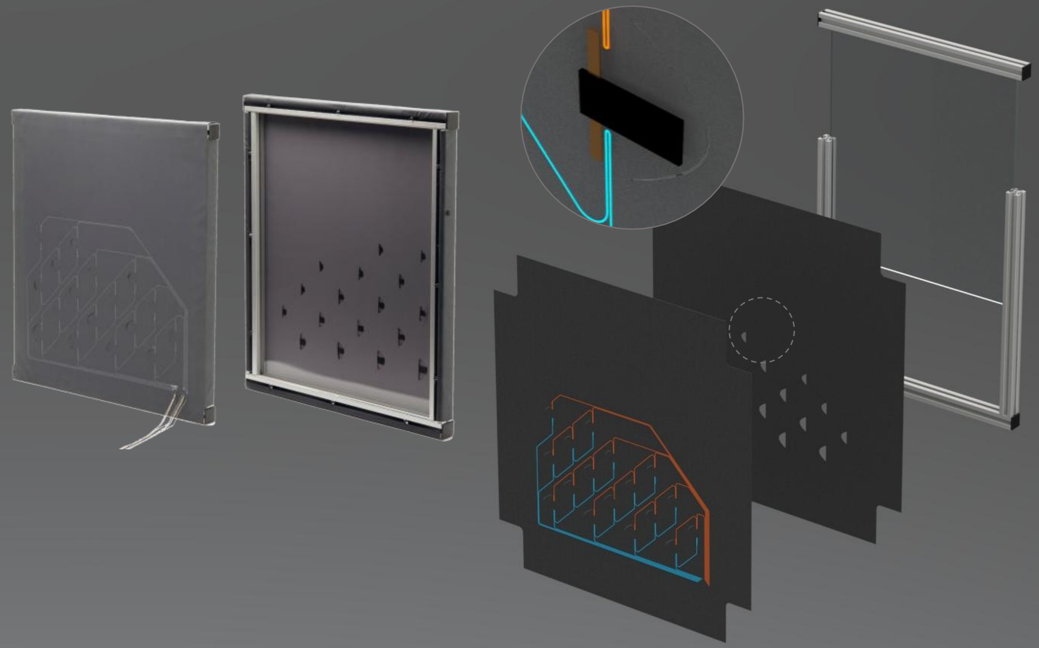
tech Integration of EAP with Textile

role Master's Thesis (2019)

2019 *University of Stuttgart (BioMat/ITKE, ICD), DE*
Fraunhofer Institute (IPA), DE
Institute of Aircraft Design (IFB), DE

task Entire Process



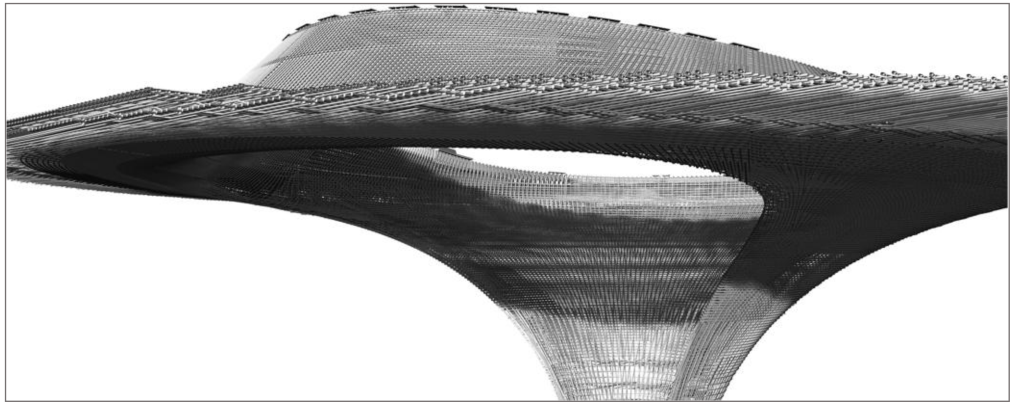


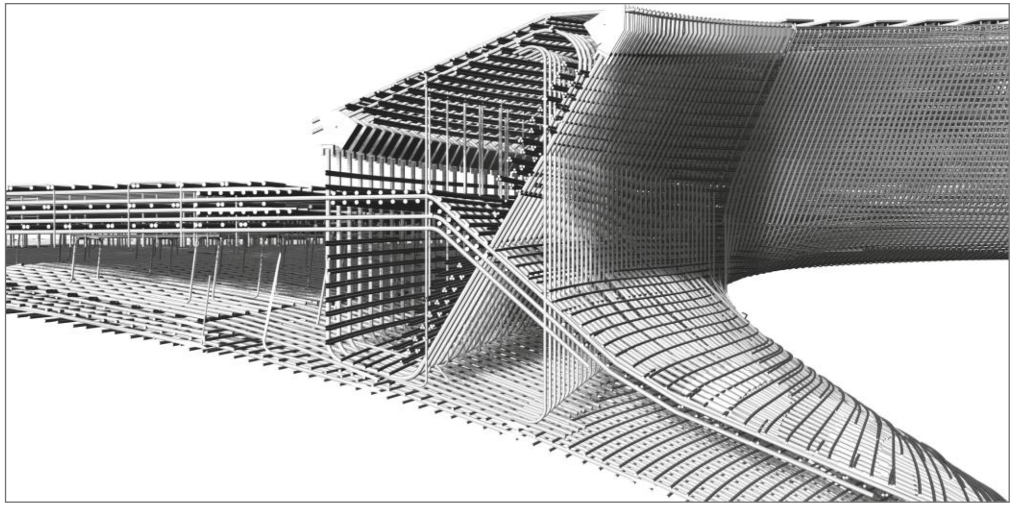
This research contributes to the ongoing SFB1244 "Adaptive Building Skins and Structures for the Built Environment of Tomorrow", funded by the German Research Foundation (DFG) under the following schema:

C03 – Electroactive polymer actuators and arrays for switchable breathability in building skins

To read more, click here.

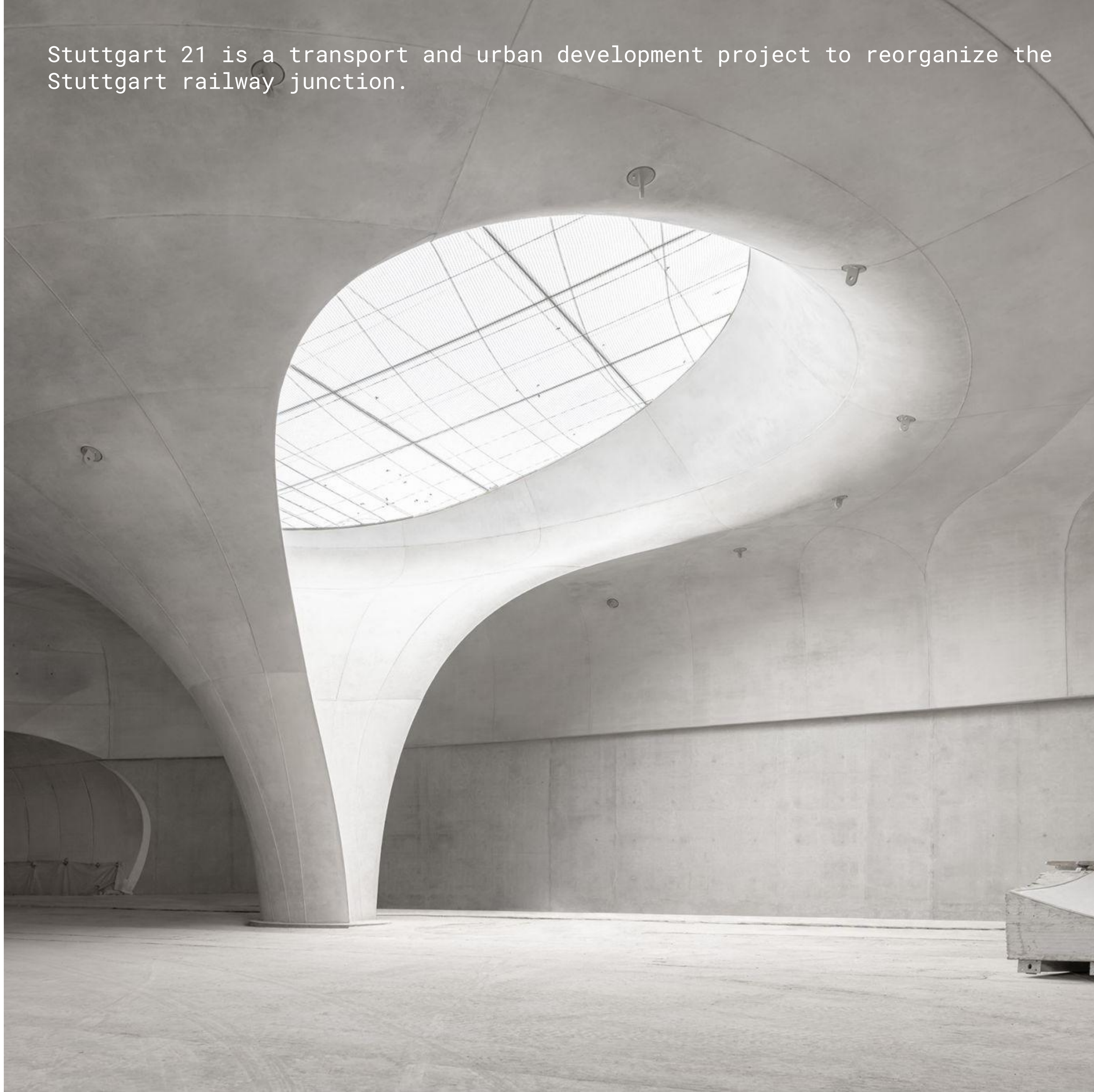
work	**MAIN STATION STUTTGART 21** Reinforced Concrete Shell
tech	Computational
role	Full-time Employment
2020	*Werner Sobek AG Stuttgart, DE*
task	Computational Design





Pictures © Werner Sobek AG

Stuttgart 21 is a transport and urban development project to reorganize the Stuttgart railway junction.



work **MODUS HIGH-RISES**
Modular Procedural Assembly

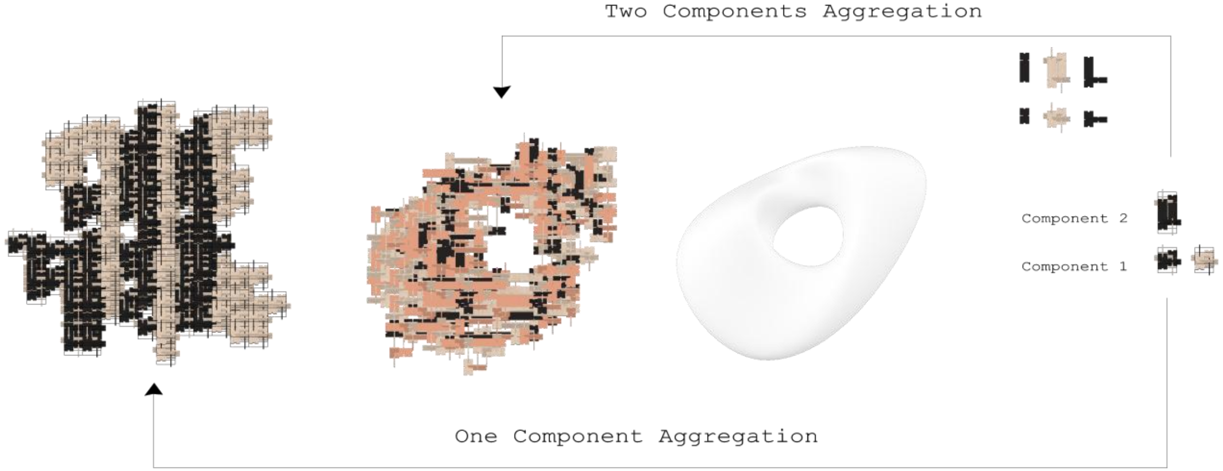
tech Computational Design

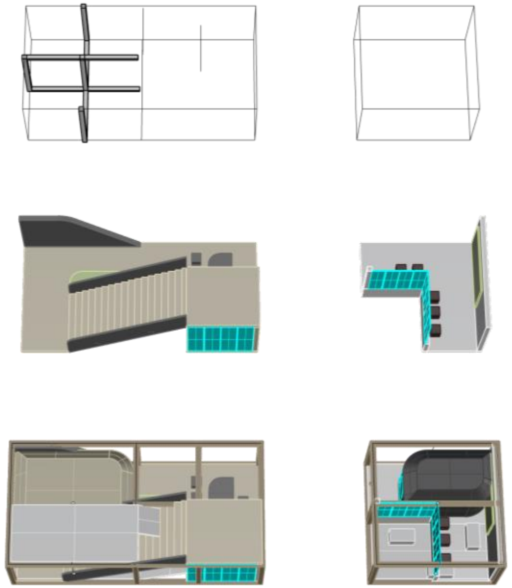
role Individual Project

2020 *DesignMorphine Workshop*

task Computation

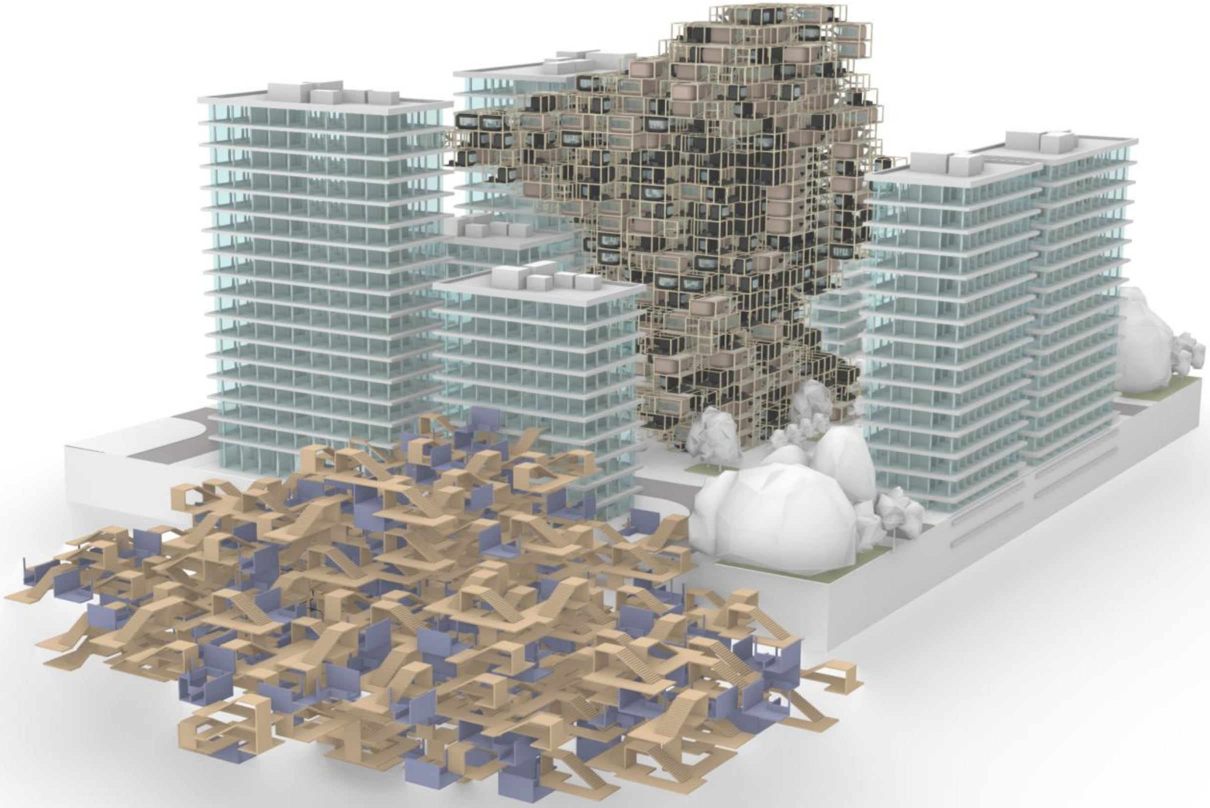
The aim is to create modular procedural modules via WASP within growing transformative spaces, that can react to the changing streams of the information society.





Modules Option 2

Modules Option 1



CHAPTER II

wood

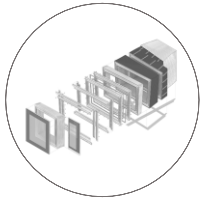
woodbased projects

The second chapter includes **selected innovative timber projects**, each highlighting one aspect of wood and its associated fabrication techniques.

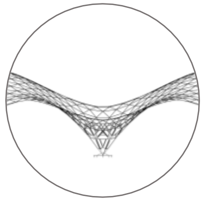
CHAPTER II

wood

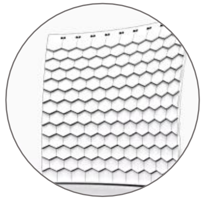
woodbased projects



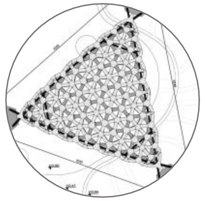
1 IBA TIMBER PROTOTYPE HOUSE
Advance CNC Fabrication



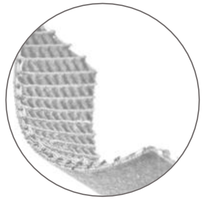
2 FUNICULAR TIMBER GRIDSHELL
Form-finding



3 SHINGLE WALL
KUKA Robotic Fabrication



4 BIOMAT PAVILION
Vacuum Molding Bio-material

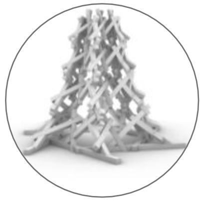


5 SMART BIO-STRUCTURE
Wood Filament 4D-Printing

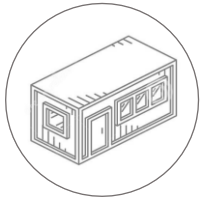
6 ODIN CHAIR
Bending Active CNC Plywood

7 HEXA TOWER
Triple Scarf Timber Joinery

8 INTER-HYGRO-JOINERY
Hygroscopic Analysis via CT Scanning



9 INTERLOCKING TOWER
Hundegger Production and AR Assembly



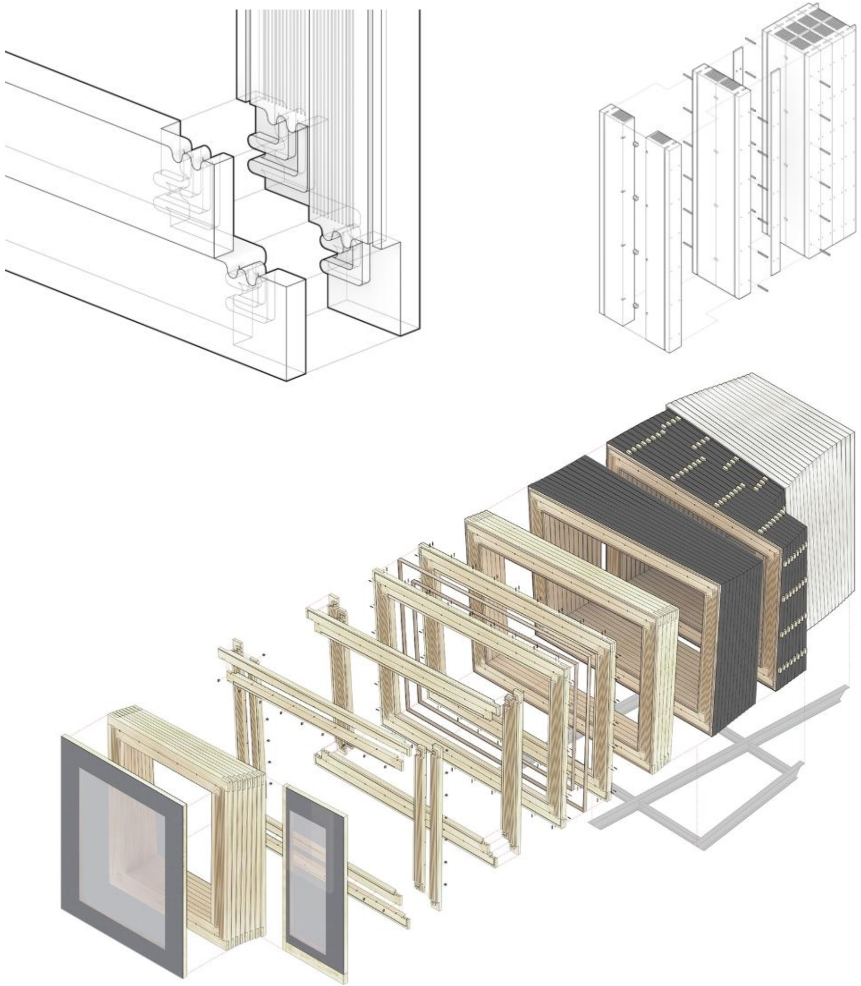
10 MODULAR HOUSING
Timber Construction and Manufacturing

work **IBA TIMBER PROTOTYPE HOUSE**
Digital Log Cabin

tech Computational Design and CNC production

role Student Assistant (HiWi)
2017 *ICD Institute, University of Stuttgart, DE*

task Construction



Pictures © ICD/ITKE University of Stuttgart

The IBA Timber Prototype House combines the benefits of traditional low-cost timber construction with advances in computational design and fabrication technologies. To read more click here

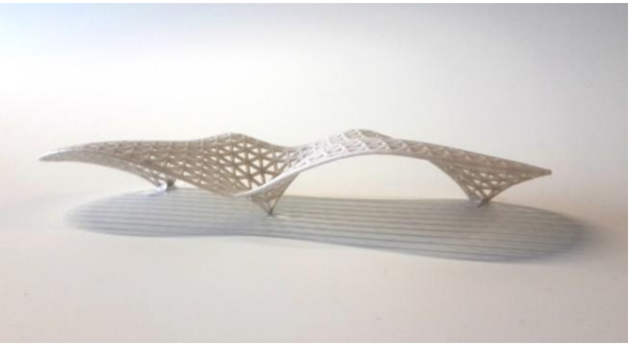
work **FUNICULAR TIMBER GRIDSHELL**
Timber Grid-shell with Steel Connectors

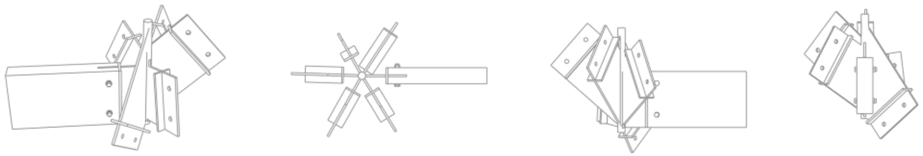
tech Form-finding and Optimization for 3D-printing

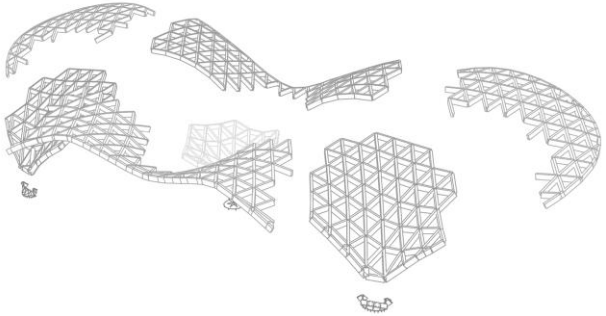
role ITECH Course: Form-finding
2017 *ITKE Institute, University of Stuttgart, DE*

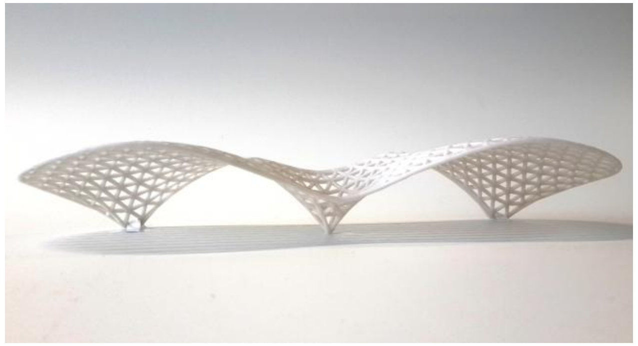
task Entire Process

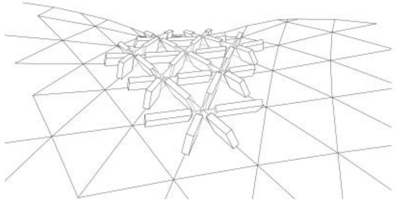
The aim of this project is to create series of bus stations as a part of VVS in Stuttgart based on funicular timber grid shell.

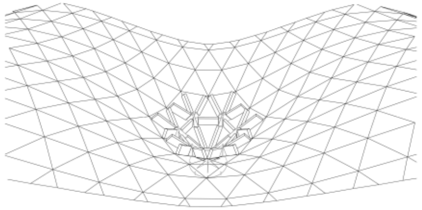


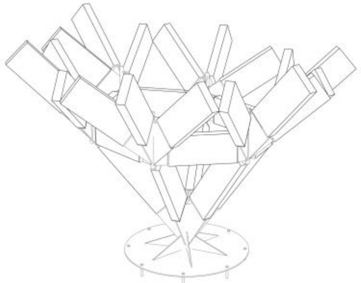


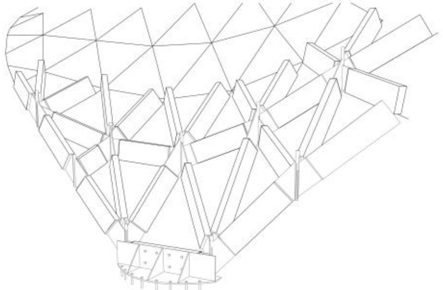




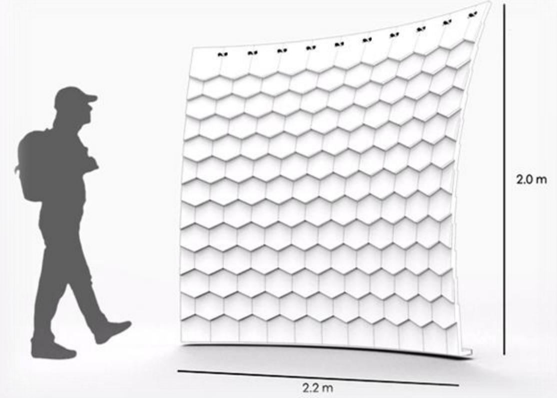


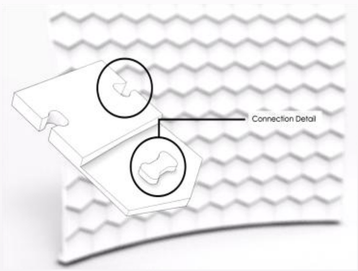


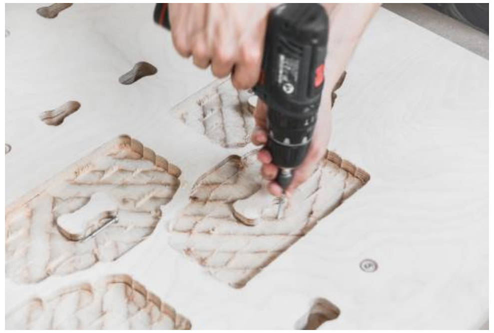




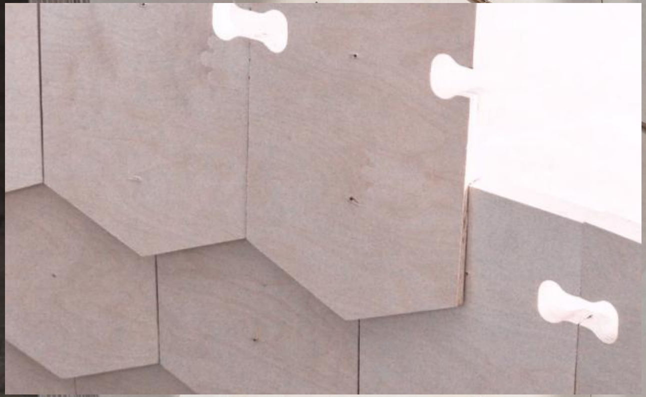
work	**SHINGLE WALL** Shingle Component
tech	Computational Design and Robotic Fabrication
role *2018*	ITECH Group Project *ICD Institute, University of Stuttgart, DE*
task	Entire Process







The Shingle Wall is a robotically fabricated segmented synclastic shell, which is the outcome of the robotic milling assignment of ITECH's Computational Design and Fabrication Seminar 2018.



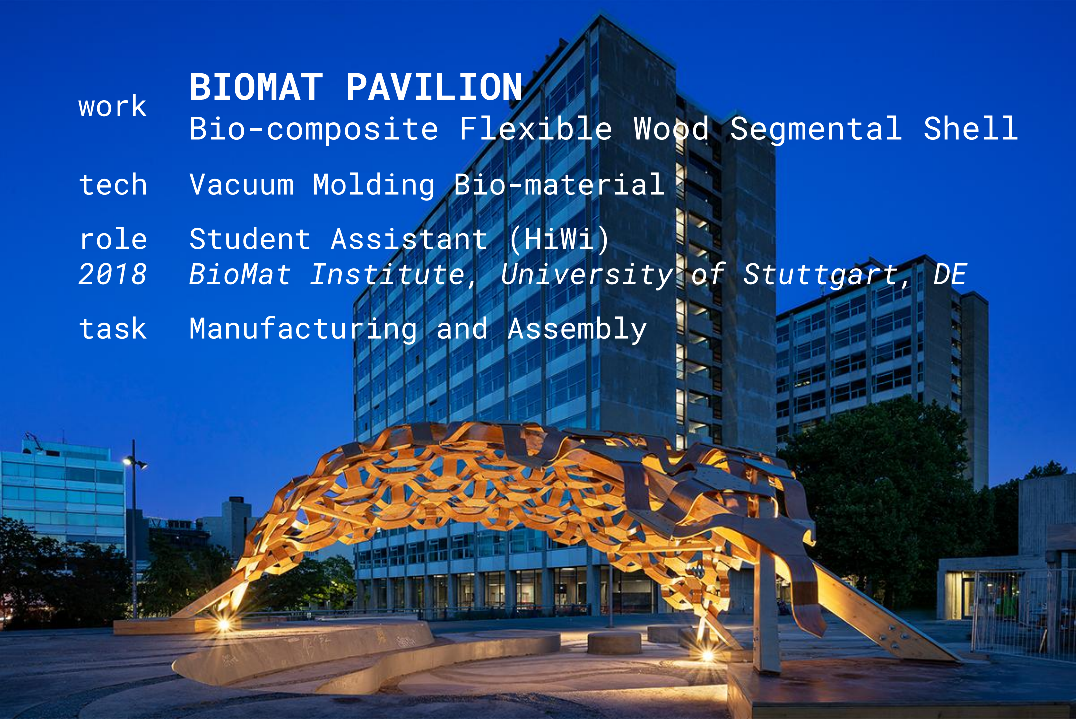
work **BIOMAT PAVILION**
Bio-composite Flexible Wood Segmental Shell

tech Vacuum Molding Bio-material

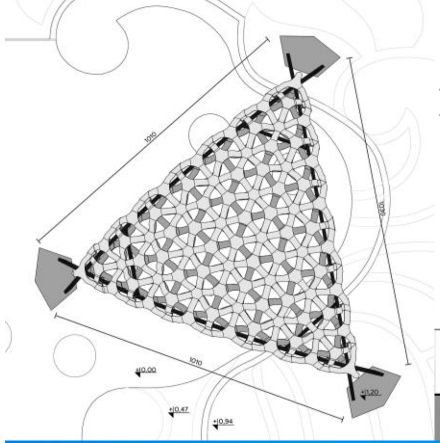
role Student Assistant (HiWi)

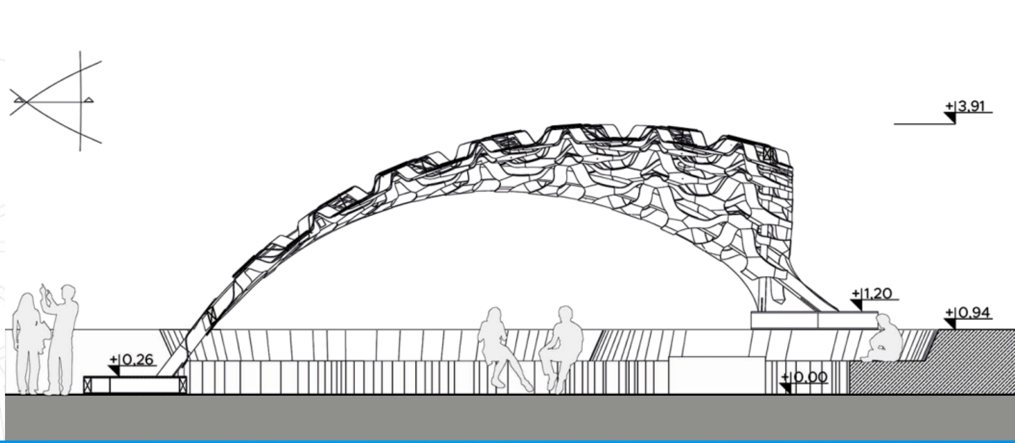
2018 *BioMat Institute, University of Stuttgart, DE*

task Manufacturing and Assembly



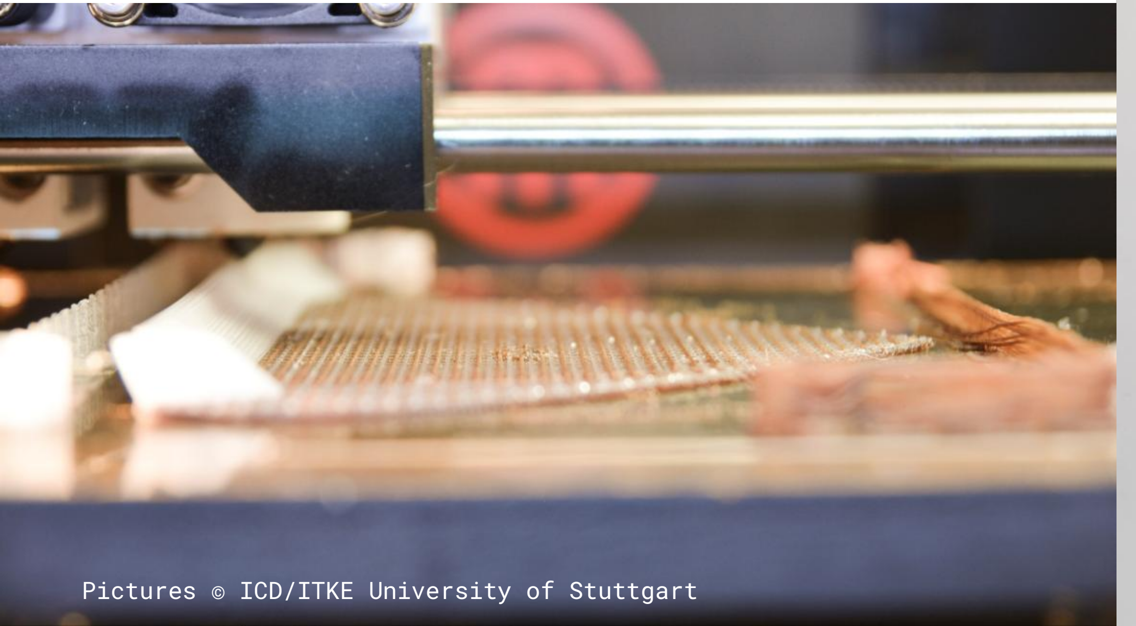
Pictures © ICD/ITKE University of Stuttgart





The pavilion is a double-curved, parametrically designed segmental shell of light, single-curved wood and bio-composite elements supported by three curved crossed wooden beams.

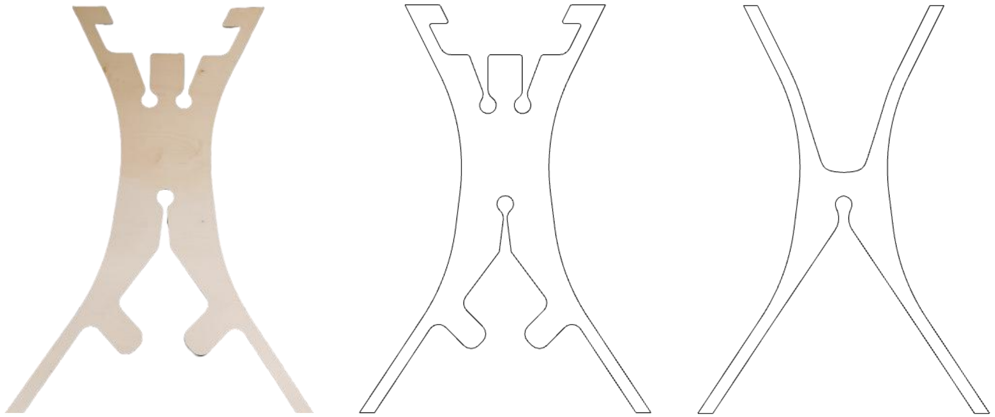
work	**SMART BIO STRUCTURE** Passive Adaptive Façade System
tech	4D Printed Programmable Hygroscopic Wood
role *2019*	Student Assistant (HiWi) *ICD Institute, University of Stuttgart, DE*
task	Simulation and Production

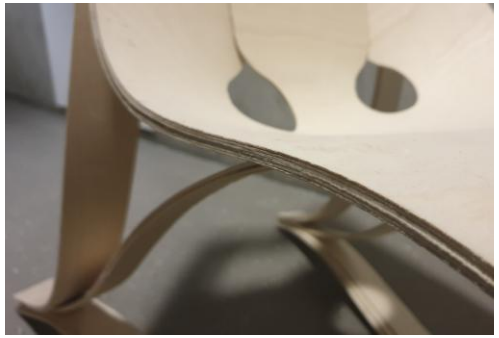


Pictures © ICD/ITKE University of Stuttgart

The potentials of programmable 4D-printing of wood filament to create environmentally passive responsive facade systems.

work	**ODIN CHAIR** Bending-active Structure
tech	Experimental Prototyping of CNC Plywood
role *2019*	ITECH Course: Chair Design *BioMat Institute, University of Stuttgart, DE*
task	Entire Process





The prepared CNC plywood layers were bended and connected in specific points leading to a 3D deformation from 2D sheets which, as a result, provided structural stiffness. To see the video, click here.

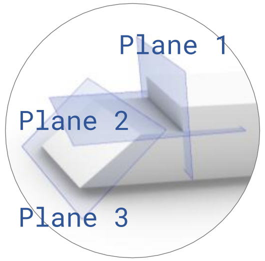
work **HEXA TOWER**
Triple Scarf Timber Joinery

tech FEA and Fabrication Method

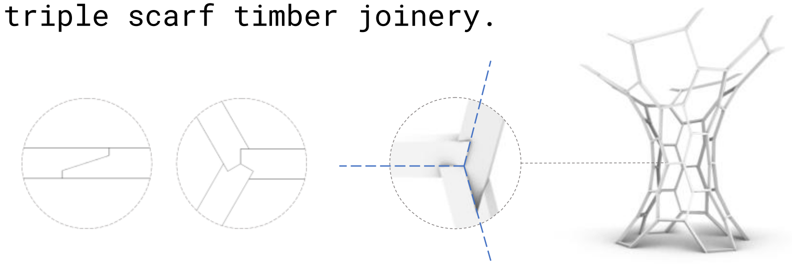
role Research Project

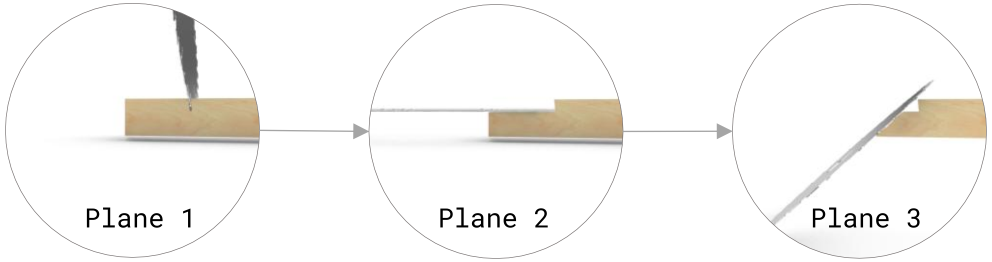
2020 *IVB Institute, NTNU i Gjøvik*

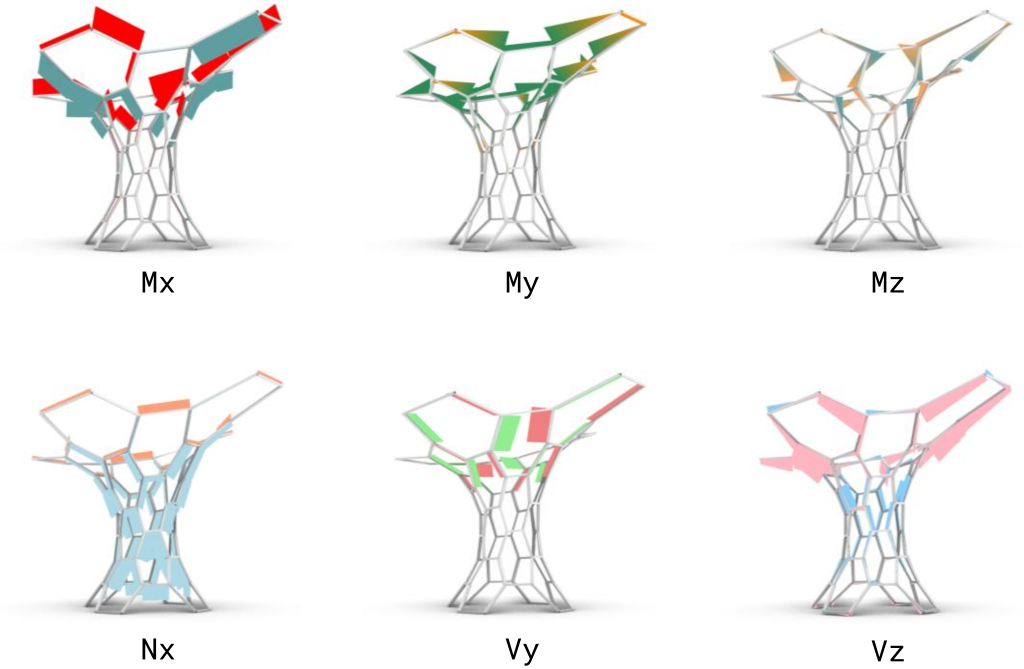
task Entire Process

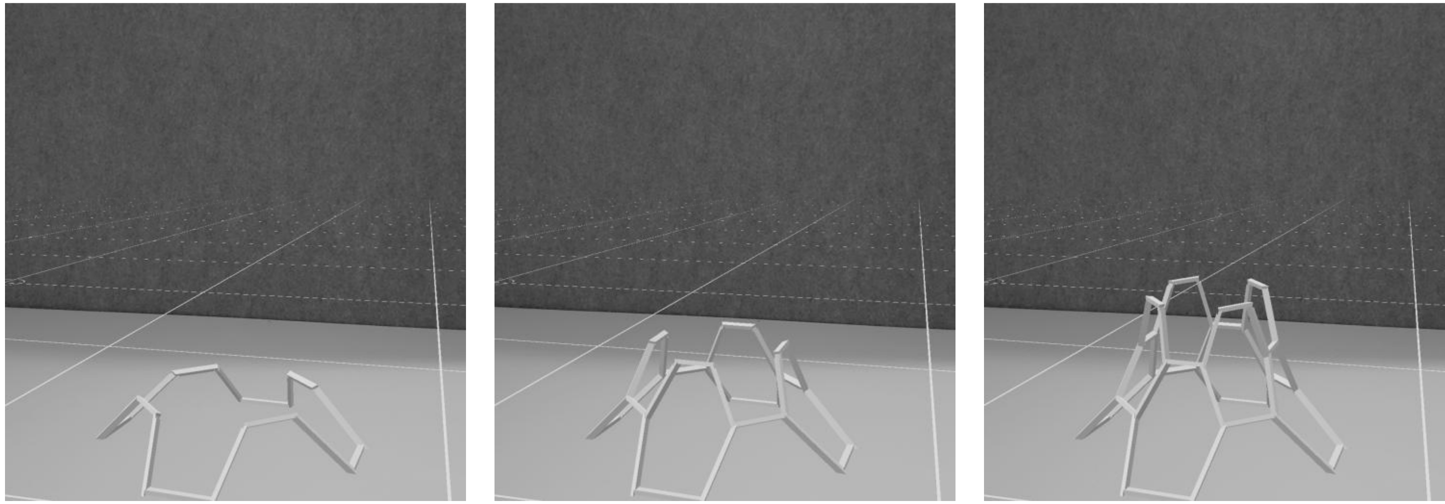


The aim of this project is to rethink of vernacular scarf joinery and extend it for triple scarf timber joinery.







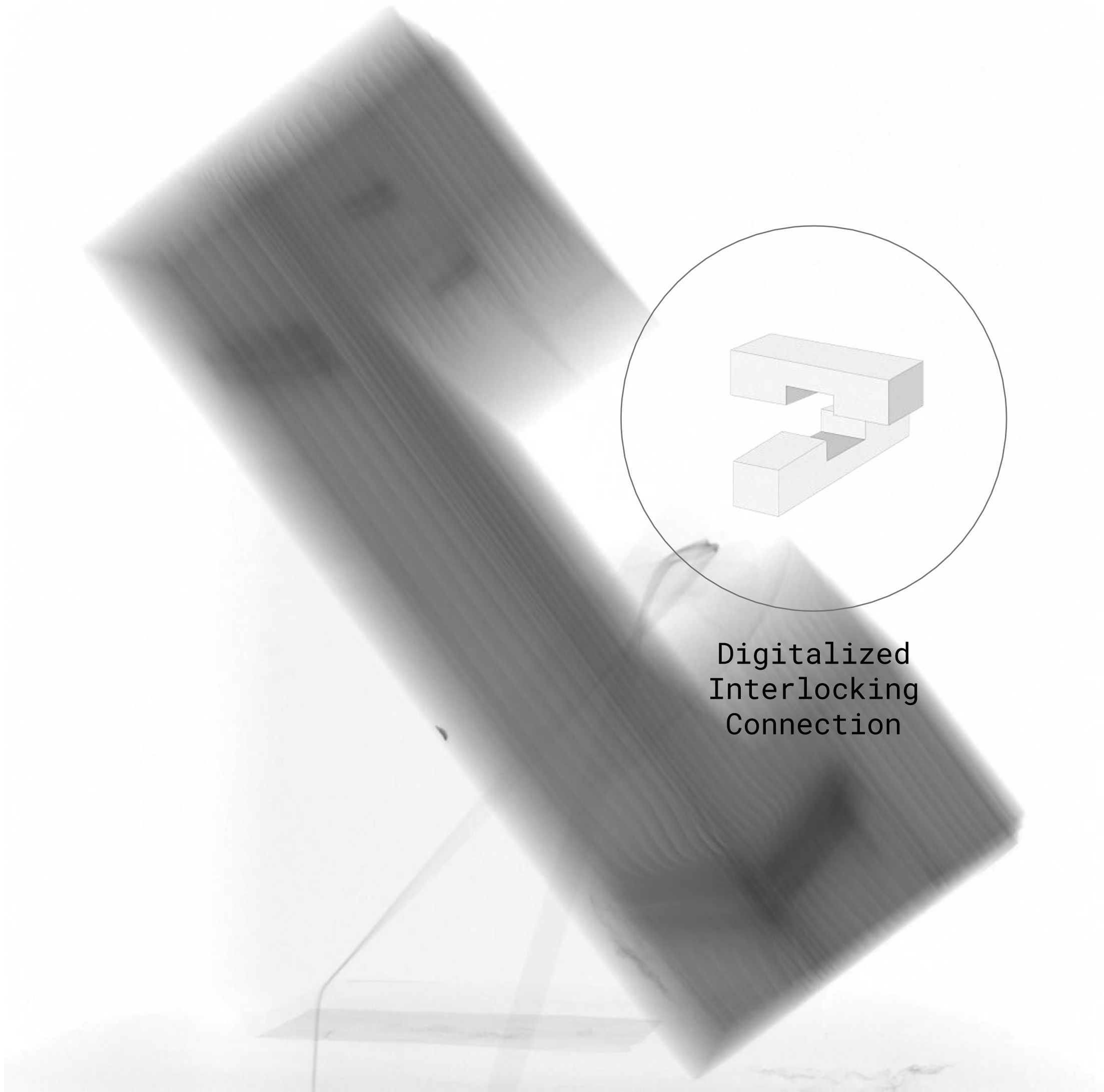


work **INTER-HYGRO-JOINERY**
Encapsulating Interlocking Timber Joinery

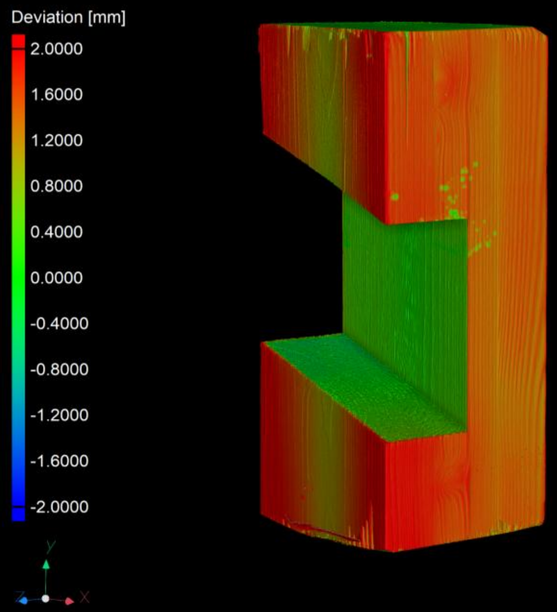
tech Hygroscopic Study of Wooden Joinery via CT Scanning

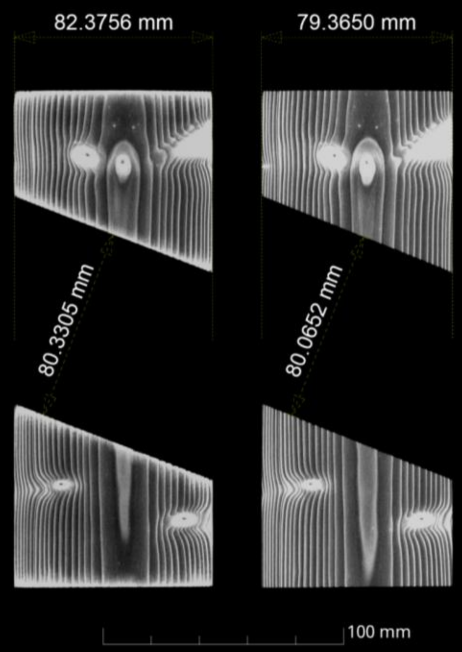
role Research Project
2021 *IVB Institute, NTNU i Gjøvik, NO*

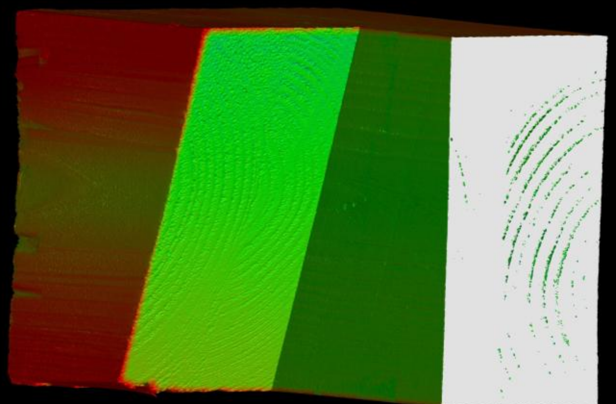
task Entire Process

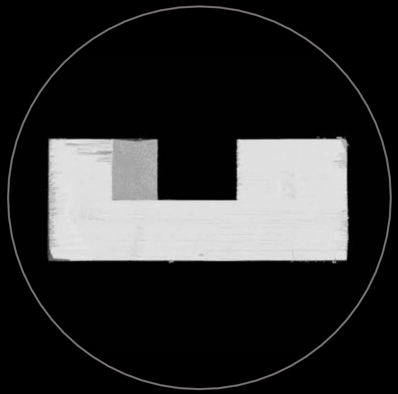


Investigation of wood's hygroscopicity via CT scanner to reinforce the structural capacity of interlocking connections.







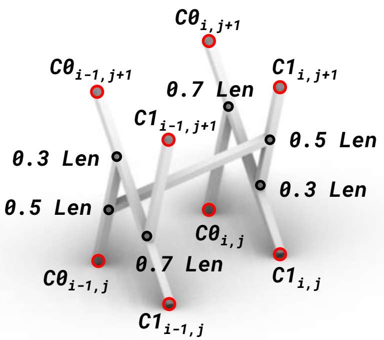


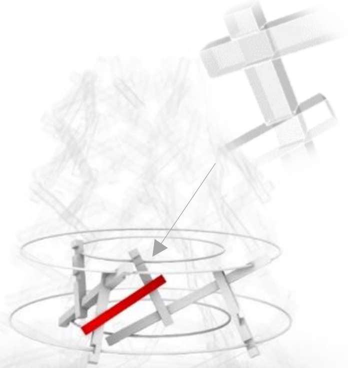
work **INTERLOCKING TOWER**
Sequential Circular Augmented Construction

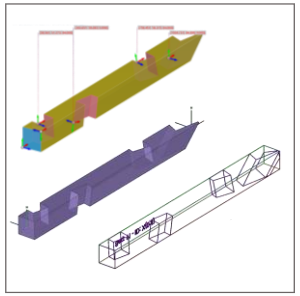
tech Hundegger Production and Sequential AR Assembly

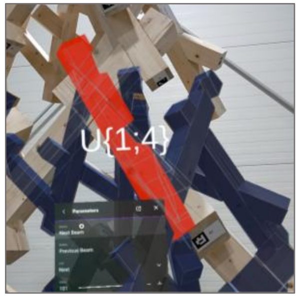
role Research Project
2022 *IVB Institute, NTNU i Gjøvik & Østlaft AG, NO*

task Entire Process









Computational Design → Hundegger Fabrication → Augmented Assembly

How innovative structures are realized through synthesizing of interlocking joinery, and how augmented-reality assembly could enable the circular economy in wood construction (?)

work **MODULAR HOUSING**
Serial Timber Housing

tech Modular Timber Construction and Production

role Full-time Employment
2023 *timpla GmbH, Eberswalde-Berlin, DE (timpla.eu)*

task Timber Construction Design

Germany's largest wooden module factory …
We are industrializing wooden modular construction to establish it as the strongest and most sustainable alternative for residential and commercial construction.